11.3 Implementing provider tests with API testing tools

For verifying contracts from provider integration tests, Postman and Dredd have been chosen to demonstrate the PactFlow BYO API testing tool functionality. PactFlow supports OpenAPI specifications and test reports to verify contract interactions. Using existing integration tests is a great way to get started with contract testing.

11.3.1 Postman and OpenAPI example

In this example, we'll use Postman to verify the contracts via the OpenAPI specification. The Newman npm package tests the endpoints using the command-line collection runner. Then, the OpenAPI specification and Newman test results for PactFlow will be published to compare with the consumer contract. Running the test from the provider project, make sure to install the Newman npm package:

```
npm install --save-dev newman
```

The provider projects that the OpenAPI specification endpoint gets downloaded as part of the package script before publishing the contract to the broker, as follows.

Listing 11.13 OpenAPI download `curl` command

```
"bi:swagger":
"curl -X GET 'http://localhost:3001/swagger.json' >
./output/movies-api.json"
```

The API tests are created with Postman as a collection of requests. The collection contains each request detailed within the swagger API specification. Export the collection to run with the Newman command-line interface (CLI), as shown in figure 11.4.